

3D Hand Contact Estimation from 2D Foundation Models

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Abstract

Estimating dense 3D contact between hands and objects from a single monocular RGB image is a core challenge in dexterous interaction understanding. We present a method for predicting binary contact maps over MANO hand meshes directly from RGB input, without access to depth or object geometry at test time. Our approach fine-tunes InteractVLM: a model that grounds vision-language semantic representations into 3D meshes, on the ARCTIC bimanual manipulation dataset, adapting it from full-body contact estimation to the hand domain. The pipeline renders MANO hand meshes into multi-view canonical images, produces 2D contact masks using a shared encoder conditioned on VLM features, and lifts these contact masks to binary per-vertex 3D contact labels via a precomputed pixel-to-vertex lookup table. The LISA-13B backbone [12] is fine-tuned with low-rank adaptation while the SAM mask decoder [10] is jointly trained to adapt from general segmentation to MANO contact prediction. Experiments on ARCTIC demonstrate the viability of adapting a full-body VLM contact estimation framework to the hand domain.

1. Introduction

Understanding how humans interact with objects is a fundamental problem in computer vision, with applications in robotics, augmented reality, virtual reality, and human-computer interaction. A key component of this understanding is contact estimation: determining which parts of the human body are physically interacting with an object. Accurate contact information provides strong geometric and semantic cues about the nature of an interaction and can improve downstream tasks such as pose estimation, motion reconstruction, and Human-Object Interaction modeling.

Recent advances in Vision-Language Models (VLMs)

have demonstrated impressive capabilities for reasoning about human-object interactions from monocular RGB images. In particular, InteractVLM [4] leverages large-scale visual and semantic knowledge to predict contact regions by grounding language-guided reasoning into canonical 3D body and object representations. However, InteractVLM was designed primarily for full-body interactions and therefore operates on coarse body-level contact representations. While suitable for interactions involving large objects such as chairs or furniture, this design limits its ability to capture the fine-grained contact patterns required for dexterous hand-object manipulation.

At the same time, the ARCTIC dataset [6] provides a unique benchmark for studying detailed hand-object interactions. Unlike previous datasets that focus primarily on rigid object grasping, ARCTIC contains dexterous bimanual manipulation of articulated objects together with accurate 3D hand and object meshes and dense contact annotations. The dataset captures challenging scenarios involving rapid motion, severe occlusions, changing object states, and highly localized contact patterns, making it particularly suitable for investigating fine-grained hand-object contact estimation.

In this work, we adapt InteractVLM for binary hand-object contact estimation. By replacing the original full-body representation with canonical MANO hand meshes and training on ARCTIC, we focus the model on fine-grained hand-object interactions. We evaluate the adapted model on allocentric and egocentric viewpoints from ARCTIC and analyze its ability to localize contact regions on interacting hands. We show that fine-tuning on a small ARCTIC subset already improves the F1 score on the egocentric view from 0.0286 to 0.0988 over the baseline, mainly through an increase in precision. The absolute score, however, remains low and is mostly limited by the training data scale.

2. Related Work

2.1. 3D Human-Object Interaction and Contact Estimation

Reconstruction of general 3D Human-Object Interaction (HOI) from monocular RGB images is a major challenge due to depth ambiguity and occlusions. Initial approaches to contact estimation relied on geometric proximity between meshes ([2] PROX-based, [22]) or sparse 2D image masks ([2] manual annotations, [16]).

More recently, Vision-Language Models (VLMs) have been introduced to bring semantic reasoning to 3D spatial tasks. One of these approaches is InteractVLM, which grounds VLM representations into 3D body models like SMPL [14] and SMPL-X [18] for contact prediction during full-body interactions.

Other recent generative approaches synthesize whole-body grasps [17], but vision-based contact localization remains difficult. The main limitation of these existing architectures is a spatial resolution mismatch. As the models are trained on large body segments, they suffer from over-activation when applied to smaller manipulations, meaning they fail to localize fine-grained contact on the hands. In order to resolve this spatial resolution mismatch, we adapt InteractVLM to use explicit 3D hand templates.

2.2. Hand-Object Interaction Datasets

Data-driven interaction modeling is largely dependent on dataset composition. Existing datasets fall into three structural categories, including body and rigid object datasets, which focus on full-body actions where the contact area is very broad. Other categories include single-hand datasets that track isolated hands performing basic and static grasps, and two-hand datasets, which capture bimanual coordination, but are often restricted to static configurations. A major limitation of all these categories is that interaction sequences are limited to static grasps, and the objects themselves are rigid.

The ARCTIC dataset tackles these limitations by providing dynamic bimanual dexterous manipulation of complex and articulated objects, such as the handling of scissors or the opening of boxes. The dataset also provides dense 3D ground-truth contact labels, which is a high quality benchmark for tracking continuous and localized contact.

2.3. Hand-Object Reconstruction and Canonical Mapping

Approaches for reconstructing 3D hand and object geometries from monocular inputs have advanced significantly. However, they are vulnerable to depth ambiguities ([3], [13]) and can yield physical inconsistencies such as disconnected configurations [8]. They furthermore make the assumption of rigid object models [5] and simple grasps [7],

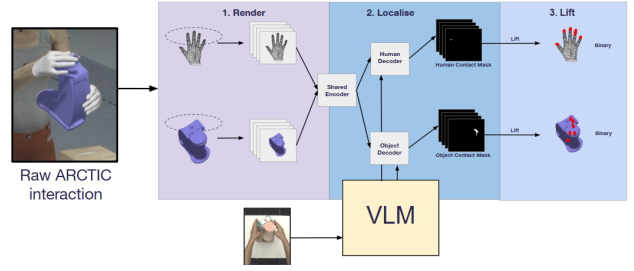


Figure 1. Overview of our architecture. **(1) Render:** MANO hand meshes are rendered into multi-view canonical images. **(2) Localise:** a shared encoder processes the rendered views; the SAM mask decoder [10], conditioned on VLM features from the raw RGB frame, produce 2D contact masks on the canonical surfaces. **(3) Lift:** the predicted masks are lifted to binary 3D contact labels on the MANO hand vertices.

meaning they are not suited to complex and articulated sequences.

In order to bypass the problems inherent to test-time depth estimation, alternative methods map interaction features into a pre-rendered 2D canonical surface [11]. While prior methods use VLMs for sparse text-to-image grounding, our approach adapts the VLM to localized and dense hand contact. By predicting dense 2D contact masks on the canonical views [1, 4], we are able to lift these into binary 3D mesh vertex coordinates using a pixel-to-vertex lookup table. This method produces strong vertex-level predictions from a single monocular image without requiring explicit depth reconstruction.

3. Methodology

3.1. Problem Description

Given an RGB image I of one or two hands interacting with an articulated object, we aim to predict a binary contact mask $\mathbf{c} \in \{0, 1\}^{778}$ over the MANO hand mesh $\mathcal{M} = (\mathcal{V}, \mathcal{F})$, where vertex v_i is labeled in-contact ($c_i = 1$) if its distance to the nearest object surface is below a threshold δ . This is done for both left and right hand meshes. The key challenge is inferring this dense 3D information from a monocular RGB image, without access to depth or object geometry at test time. We approach this by finetuning InteractVLM, which was originally designed for predicting full-body contact. By training it on the ARCTIC dataset, we adapt it to the hand domain by substituting MANO for the body mesh (SMPL). We train and evaluate under the ego-centric viewpoint I_{ego} ; the baseline of 4.1.1 also reports a frontal allocentric I_{front} result.

3.2. InteractVLM

To extract semantic information, we utilize InteractVLM as our backbone model. In our implementation, the input im-

age I is taken from the egocentric viewpoint I_{ego} provided by ARCTIC.

The pipeline operates in three stages. First, the MANO hand mesh is rendered into four multi-view canonical images, which are encoded by the SAM image encoder [10]. The raw RGB interaction frame is processed in parallel by a pretrained CLIP vision transformer [19], whose features condition LISA-13B [12], a language-vision model that outputs a $\langle \text{SEG} \rangle$ token that contains the model’s semantic interpretation of where the contact occurs. Second, the SAM mask decoder consumes the canonical-view embeddings together with the $\langle \text{SEG} \rangle$ token embedding to predict a 2D contact mask for each canonical view. Third, these 2D masks are lifted back to binary 3D contact labels on the underlying MANO vertices through the lookup table as described in 3.3. By rendering into a canonical pose before localization, the model sidesteps the need for depth estimation, operating entirely in a geometry-aware representation space. An overview of our architecture is shown in Figure 1.

3.3. 3D Mesh Contact Classification

During the localization stage, the VLM produces a $\langle \text{SEG} \rangle$ token whose embedding is used by the SAM mask decoder [10] to generate four 2D contact masks, one per canonical view of the hand mesh. To get a binary contact classification on the vertices of the 3D MANO hand mesh a precomputed pixel-to-vertex lookup table is used. For every pixel in the rendered views, this table indicates which vertex on the surface that pixel belongs to.

A single canonical mesh is created by combining the right and left MANO templates into one mesh of 1556 vertices and 3076 faces, with an offset of 0.15m between the hands to prevent occlusion. Using PyTorch3D [21] the mesh is then rendered from four fixed viewpoints; palm, back, left and right, at 1024×1024 resolution, then filling in each pixel by the local surface normal mapped to RGB, encoding the orientation into the color. For each viewpoint the lookup table stores, for every pixel, the three vertex indices of the underlying triangle together with their barycentric weights. Since the mesh and cameras are fixed, the lookup table is computed once at the beginning and then reused for all frames.

Each vertex is visible from several of the four views. A MANO vertex is labeled as in-contact (1) if at least one of these views has a predicted mask covering a pixel that corresponds to that vertex through the lookup table and not-in-contact (0) otherwise. These per-vertex labels are then evaluated against the ground truth. The same lookup table is used in the opposite direction to create the training targets: the 3D ground truth contact vertices from ARCTIC are mapped through it to mark the corresponding pixels, producing a supervised 2D mask for each view.

3.4. Training

For the training we use the `hcontact` configuration from InteractVLM, modifying only the dataset and target mesh. A new `arctic.hcontact` configuration is added which makes the training loop use ARCTIC sequences and the combined MANO lookup table from 3.3, instead of DAMON [22] and SMPL.

Following the architecture as described in 3.2, LISA-13B [12] is fine-tuned with low-rank adaptation ($r = 8$, $\alpha = 16$) [9]. The CLIP vision transformer [19] and the SAM image encoder [10] are kept frozen, while the LoRA adapters and the SAM mask decoder are trained jointly. This way the decoder adapts from general object segmentation to MANO contact masks.

The loss is calculated by taking the weighted sum of $1.0 \times$ cross-entropy on the VLM text output, $2.0 \times$ binary cross-entropy, $1.0 \times$ Dice loss on the 2D contact masks, and a second $3.0 \times$ binary cross-entropy on the lifted 3D vertex predictions.

4. Experiments

4.1. Dataset and Setup

We train and evaluate our method on the ARCTIC dataset [6], which provides multi-view RGB video of two hands dexterously manipulating 11 articulated objects, captured from eight allocentric viewpoints and one egocentric viewpoint. Each frame includes 3D MANO hand and object meshes, used to compute the ground-truth binary contact labels.

4.1.1. Baseline

To establish a baseline, we ran InteractVLM in inference mode using the `hcontact` configuration, which performs 3D human contact estimation. We evaluated the model on an ARCTIC sequence *box_use*.

Each sequence was preprocessed to extract per-frame inputs. Ground truth contact labels were derived using a threshold of **3 mm** between MANO hand vertices and the object surface. For InteractVLM, the default contact probability threshold of 0.3 was used for the primary evaluation.

InteractVLM operates internally on an SMPL body model that is subsequently lifted to SMPL-X. To compare predictions against MANO ground-truth vertices, we identified the SMPL-X to MANO vertex correspondence (MANO is a direct subset of SMPL-X) and applied this index mask to extract only the hand vertices from the predicted contact scores, bringing both representations into a common MANO space. Precision, recall, and F1 score were then computed over all frames. For the fine-tuned model we also report gIoU and cIoU of the predicted 2D contact masks, averaged over the four canonical views.

Two viewpoints were evaluated: an *allocentric* (top-frontal) view and an *egocentric* view. Beyond the default configuration, we probed sensitivity to several hyperparameters. Prediction thresholds ranging from 0.1 to 0.9 were swept, and the input prompt was modified from the default:

“Which body parts are in-contact with the {object}? Segment these contact areas.”

to a hand-specific variant:

“Which areas of the human hands are touching or gripping the {object}? Segment the contact regions.”

4.1.2. Training

For training we preprocess ARCTIC into the per-frame format expected by the `hcontact` configuration of InteractVLM. We use the official `process_seqs.py` script to get the hand (MANO) and object meshes for every frame, with the correct pose and position already applied. The script was modified first, since it expects MANO to return 21 joints, while the original MANO only outputs 16. We fix this by appending the five fingertip vertices to the output. For every frame we mark a MANO vertex as in-contact if it lies within **3 mm** of the nearest object vertex, as defined by ARCTIC [6]. For each frame and each of the four canonical views the resulting contact vertices are projected through the lookup table of 3.3 with PyTorch3D [21] to render the ground-truth 2D mask targets.

Since training on the full ARCTIC dataset is too expensive in both time and compute, we use a subset of 1720 frames, taken from three subjects (s01, s02, s06), split randomly into a 1376 training and 344 validation set (80/20). The RGB input is the egocentric camera (view 0), which matches the egocentric setting used for the baseline in 4.1.1. The new `arctic_hcontact` dataset loader added to the InteractVLM code base reads the preprocessed frames, lookup table and contact labels and feeds them into the original training loop unchanged.

Implementation Details We start from the public LISA-13B [12] checkpoint and train with AdamW [15] under DeepSpeed ZeRO-2 [20] in bfloat16, using batch size of 8, learning rate of $3 \cdot 10^{-4}$ with cosine annealing and 30 epochs of 500 steps. We build on the original InteractVLM implementation and only add the `arctic_hcontact` configuration and the ARCTIC data loader as described in 4.1.2. The training was run in parallel on two NVIDIA A100 GPUs.

4.2. Quantitative Results

4.2.1. Baseline

The quantitative results for the baseline evaluation are reported in Table 1. The model exhibits a consistent pattern

Table 1. Contact prediction performance on the ARCTIC validation set, with a contact threshold of 0.3 for both models. *Baseline*: pretrained InteractVLM without fine-tuning. *Ours*: fine-tuned on ARCTIC.

View	Method	Precision	Recall	F1
Frontal	Baseline	0.0142	0.9422	0.0280
	Ours	—	—	—
Egocentric	Baseline	0.0145	0.9567	0.0286
	Ours	0.0753	0.1505	0.0988

across both viewpoints: very high recall with an extremely low precision. For the allocentric view, the combined F1 score across both hands is 0.0280. For the egocentric view, the combined F1 score is 0.0286.

This behaviour stems from the model predicting contact for a large fraction of hand vertices in almost every frame. Ground-truth contact regions typically span 10–50 vertices per hand out of 778 total MANO vertices, whereas InteractVLM consistently labels 600–650 vertices as contacts, producing the large false positive counts visible in Table 1.

Varying the probability threshold of InteractVLM, modifying the prompt, and switching between ARCTIC sequences did not yield any meaningful improvement. Results remained largely stable across all configurations, indicating that the model’s tendency to over-predict contact is not sensitive to these parameters.

4.2.2. Training

Table 1 shows the contact prediction results of the fine-tuned model. It can be observed that for the egocentric view the precision increased from the baseline 0.0145 to 0.0753, and the F1 score from 0.0286 to 0.0988. Recall dropped from 0.9567 to 0.1505. Further, the frontal view was not evaluated, since the training was only done on the egocentric view.

The decreasing recall and increasing precision show that fine-tuning has reversed the baseline’s behaviour: while the baseline labels 600–650 vertices, our model averages to 39 vertices as contacts per frame across the validation set, indicating that our model is more conservative in its prediction than before. Both the baseline and finetuned models still have low precision in absolute terms, but the baseline activates too many vertices whereas the fine-tuned model activates only a small fraction of them.

The 2D contact masks themselves are predicted accurately on the validation set; we observe a gIoU of 0.885 and a cIoU of 0.938 across all canonical views, showing that the localization of contact regions on the canonical mesh is mainly correct. The much smaller per-vertex F1 score therefore seems to come somewhere later in the pipeline, possibly when lifting from 2D to 3D, where masks with

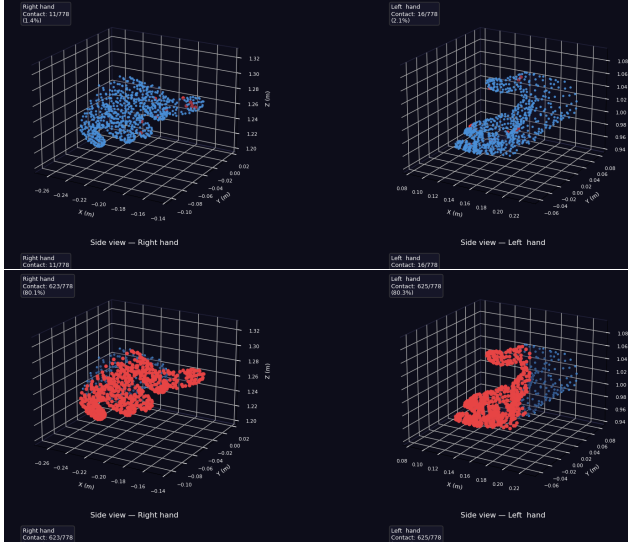


Figure 2. MANO contact visualisation for frame 200 (frontal view). *Top*: ground truth (11/778 R, 16/778 L contact vertices). *Bottom*: baseline prediction at threshold 0.3 (623/778 R, 625/778 L). (Red = Contact, blue = No contact)

only a few contact pixels translate into a small number of active MANO vertices.

4.3. Qualitative Results

4.3.1. Baseline

When comparing ground-truth and predicted contact maps for the same frame, the ground truth shows a small cluster of highlighted contact vertices concentrated at a region, while the InteractVLM baseline colours nearly the entire hand surface as contact. This overactivation is consistent across both the frontal and egocentric viewpoints and across both evaluated sequences.

4.3.2. Training

Figure 3 shows the predicted MANO contact for frame 201 next to the ground truth. Unlike the baseline, which marks nearly the whole hand surface as in contact, the fine-tuned model produces sparse predictions mostly around the fingertips. The predicted counts (10/778 right, 11/778 left) are close to the ground-truth counts (10/778 and 15/778), and the activated vertices sit in roughly the same area as the ground-truth contact region.

These results match the quantitative results in 4.2.2: the model no longer over-predicts the hand surface, but it still misses part of the true contact region, which also shows up in the low recall.

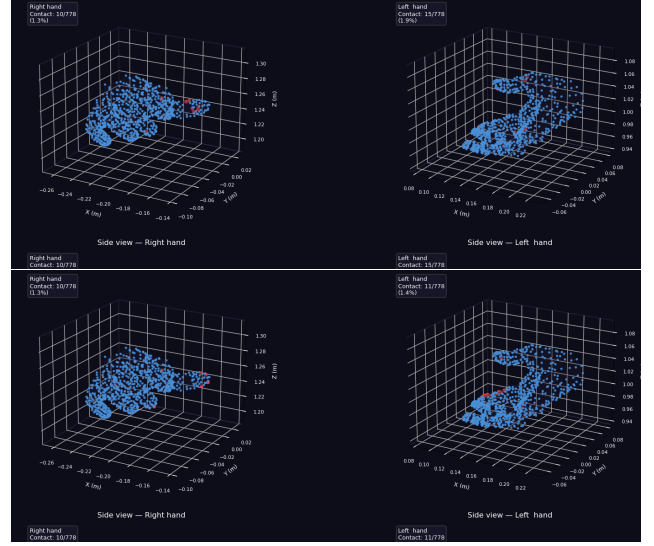


Figure 3. MANO contact visualisation for frame 201 (egocentric view). *Top*: ground truth (10/778 R, 15/778 L contact vertices). *Bottom*: fine-tuned model prediction at threshold 0.3 (10/778 R, 11/778 L). (Red = Contact, blue = No contact)

4.4. Discussion

4.4.1. Baseline

The baseline results reveal a fundamental mismatch between what InteractVLM was trained to predict and the fine-grained contact localization required by ARCTIC. The model was designed and trained on datasets where contact regions span large body surface areas like torso, arms, and legs interacting with furniture or large objects. Applied to dexterous hand manipulation, where contact is confined to a small number of fingertip and palm vertices, the model has no mechanism to suppress its tendency toward broad activation.

The near-perfect recall (>0.93 across all conditions) confirms that the model does capture the presence of hand-object contact, but it simply cannot localize it. The insensitivity to threshold variation, prompt engineering, and sequence choice further suggests that this is a structural limitation of the pretrained model rather than a tuning issue. Fine-tuning on ARCTIC data is therefore a necessary step, not merely an optional improvement.

4.4.2. Training

Fine-tuning on the ARCTIC dataset shifts the model from the overpredicting behaviour of the baseline towards more conservative predictions. The high 2D mask scores ($\text{gIoU} = 0.885$, $\text{cIoU} = 0.938$) suggest that the visual reasoning has mainly adapted to MANO, while the per-vertex F1 score remains low in absolute terms despite improving over the baseline. This indicates that the issue lies in the

per-vertex prediction step rather than in how the model sees the image.

The most likely cause is the size of the training set. Due to limited compute and time constraints we fine-tuned on approximately 0.8% of ARCTIC, and the training loss was still decreasing at the final epoch, suggesting that the model had not yet converged. Future work could focus on scaling the training data to improve the per-vertex results.

Other changes to improve the per-vertex predictions could involve reweighting the loss function to take the class imbalance into account. We also keep the decision threshold on the predicted contact scores at the value of 0.3 used by the baseline. Adjusting this threshold could trade some precision for higher recall. Another limitation of this experiment is that the geodesic distance metric reported in the InteractVLM evaluation was not included, because it requires a precomputed SMPL distance matrix that does not transfer to MANO.

5. Conclusion, limitations and author contributions

We adapted InteractVLM from full-body contact estimation on SMPL to hand-object contact estimation on MANO by fine-tuning the model on the ARCTIC dataset. The fine-tuned model becomes more conservative in its predictions, unlike the baseline, which tends to over-predict at the same threshold of 0.3. The F1 score on the egocentric view improves from 0.0286 to 0.0988, mostly through an increase in precision. The high 2D mask scores ($gIoU = 0.885$, $cIoU = 0.938$) also indicate that the visual reasoning has mainly adapted to the MANO geometry.

The per-vertex F1 score is still low in absolute terms, mainly because our limited compute and time allowed us to train on only a small fraction of ARCTIC (0.8%). Future work could explore increasing the amount of training data, reweighting the loss to account for class imbalance, adjusting the decision threshold, and computing a MANO-specific geodesic distance matrix. Our results indicate that the InteractVLM pipeline can be adapted to MANO, with training data scale and threshold calibration as the main directions for further improvement.

The work for this project was distributed evenly over all six co-authors.

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